



UNIVERSITY OF BUEA

Faculty of Engineering and Technology

CEF440

Internet Programming and Mobile Programming

Group 16

TASK 6: DATABASE AND BACKEND DEVELOPMENT

Submitted to:

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TASK 6: DATABASE AND BACKEND DEVELOPMENT FOR NETINSIGHT

6.1 Introduction

The NetInsight application was designed to collect, store, and analyze anonymous network performance data from mobile users. To achieve this, a database system and backend service were developed to manage network measurement records and user feedback. The backend exposes RESTful API endpoints that allow the mobile application to communicate with the database efficiently and securely.

6.2 Identification of Data Elements

The first step in database design is identifying the data elements required by the application.

A. Network Metrics Data (NetworkReading)

The system stores network performance measurements collected from users' devices. The main data elements include: reading_id, timestamp, latitude, longitude, network_type, latency, etc for network readings.

B. User Feedback Data (UserFeedback)

The system also stores anonymous user feedback regarding network experience. Data elements include: feedback_id, rating, note, linked_reading_id etc, for user feedback

6.3 Conceptual Database Design

The conceptual design was developed to identify the major entities and their relationships without considering implementation details.

Main Entities

1. NetworkReading

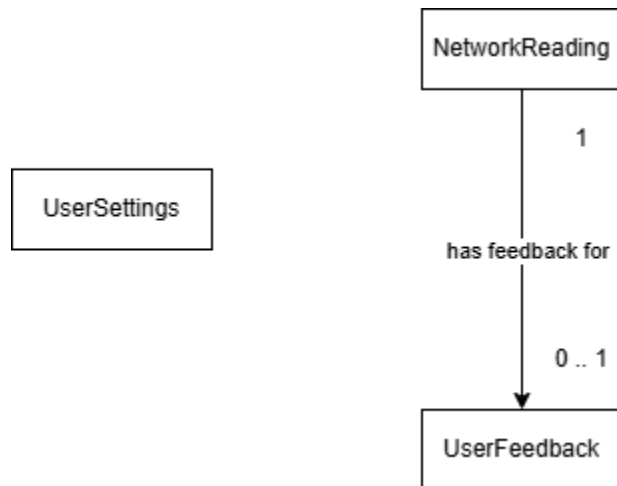
Stores network measurement information collected by the application.

2. UserFeedback

Stores user feedback and ratings concerning network quality.

3. UserSettings

Stores user preferences.



6.4 Entity Relationship (ER) Diagram

The conceptual model was transformed into an Entity Relationship (ER) Diagram to provide a structured representation of the database.

Entities:

Network Metric (NetworkReading)

Reading_id(PK)

timestamp

networkType

signalStrength

downloadSpeed

...

Feedback (UserFeedback)

feedbackId (PK)

rating

note

linked_reading_id (FK)

...

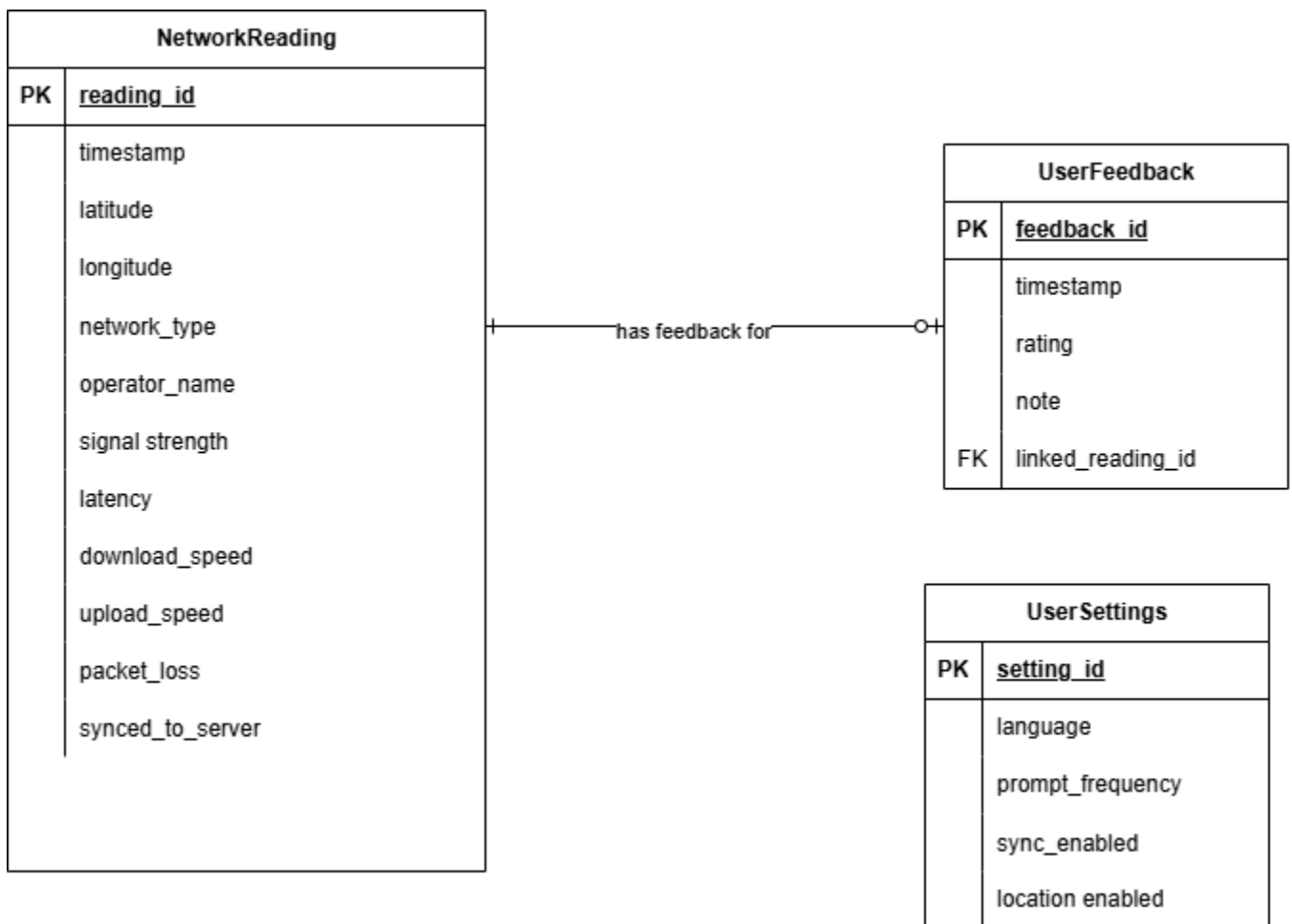
Settings (UserSetting)

Setting_id(PK)

Language

Prompt_frequency

The ER diagram illustrates the attributes associated with each entity and serves as the blueprint for database implementation.



6.5 Database Implementation

Database Technology

MongoDB Atlas was selected as the database management system due to its:

- Flexible document-oriented structure.
- Scalability.
- Cloud-hosted availability.
- Easy integration with Node.js applications.

Database Collections

The implementation resulted in two main collections:

NetworkMetrics Collection

Stores all network performance measurements received from the mobile application.

An example of the type of data it collects:

For network metric (NetworkReading):

```
{ "deviceId": "device-ios", "timestamp": "2026-06-12T23:40:04.599Z", "connectionType":  
"cellular", "isConnected": true, "isInternetReachable": true, "signalStrength": -100,  
"downloadSpeed": 2.08, "downloadStatus": "fair", "uploadSpeed": 0.1, "uploadStatus": "poor",  
"latency": 198 }
```

For feedback (UserFeedback) collection

```
{ "experience": "good", "areaOfFeedback": "data_speed", "note": "Slow downloads in the  
evening", "rating": 4, "networkProvider": "Anonymous", "resolved": false }
```

MongoDB Atlas Configuration

The database was hosted on MongoDB Atlas using a cloud cluster. Database access credentials and connection strings were configured through environment variables to enhance security.

6.6 Backend Implementation

Backend Technology Stack

The backend was developed using:

- Node.js
- Express.js
- MongoDB Atlas
- Mongoose ODM
- railway

Express.js was used to create RESTful APIs, while Mongoose provided object modeling and communication between the application and MongoDB.

API Endpoints

Network Metrics APIs

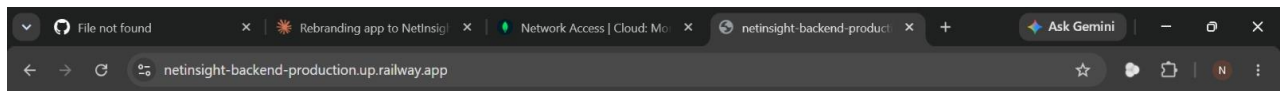
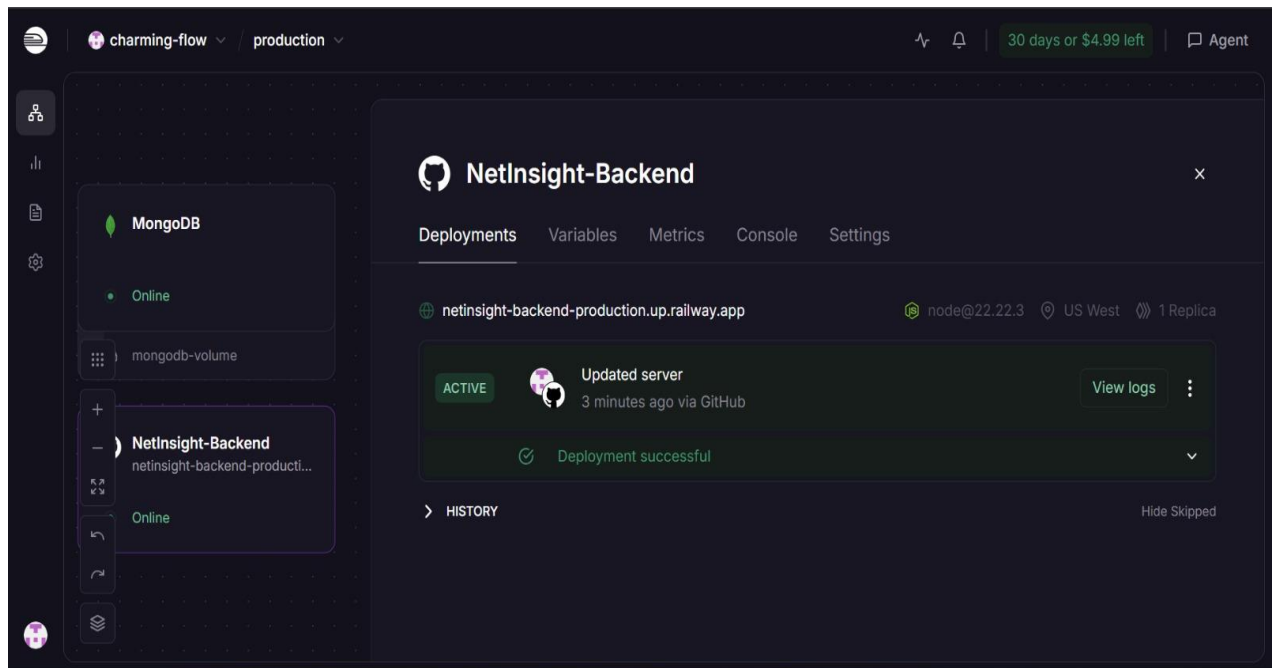
Method	Endpoint	Function
GET	/	Health check
POST	/api/network-metrics	Save network measurements
GET	/api/network-metrics	Retrieve recent measurements
GET	/api/network-metrics/stats	Retrieve network statistics

Feedback APIs

Method	Endpoint	Function
POST	/api/feedback	Submit feedback
GET	/api/feedback	Retrieve feedback
PATCH	/api/feedback/	Update feedback resolution status

Backend Features

- Stores anonymous network performance measurements.
- Stores user feedback and ratings.
- Computes network statistics.
- Retrieves historical measurement data.
- Supports feedback management and resolution tracking.
- Provides a health-check endpoint for monitoring service availability.



NetInsight backend running...

6.7 Connecting the Database to the Backend

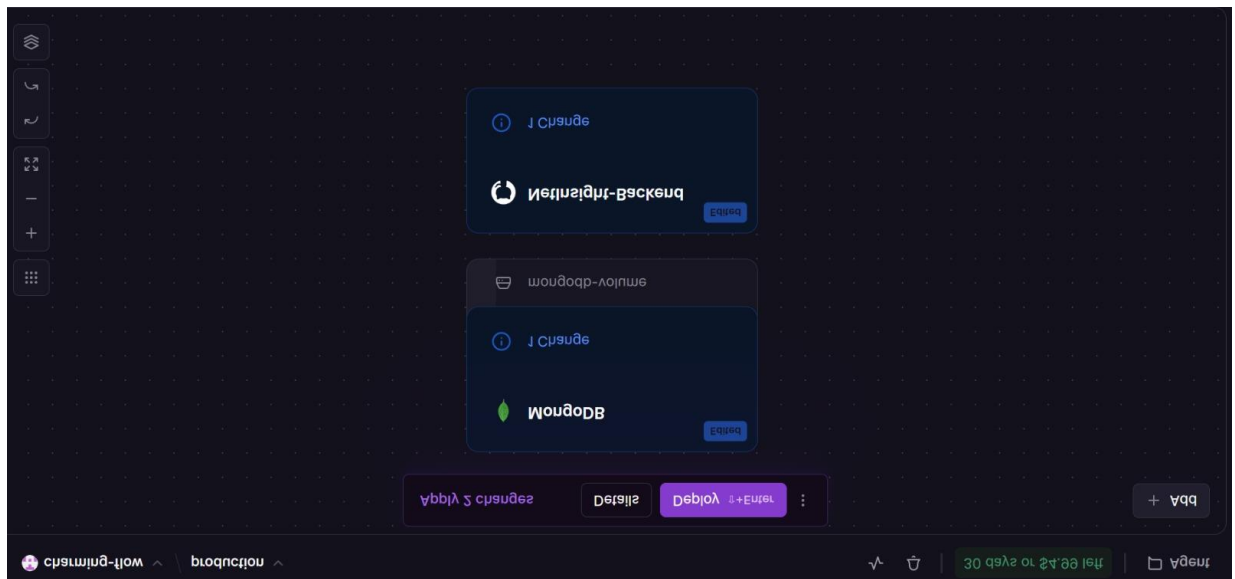
The backend communicates with MongoDB Atlas using Mongoose and a MongoDB connection string stored in an environment variable.

Connection Process

1. Create a MongoDB Atlas cluster.
2. Create a database user and password.
3. Configure network access permissions.
4. Obtain the MongoDB connection string.
5. Store the connection string in a .env file as:
`MONGODB_URI=mongodb+srv://username:password@cluster.mongodb.net/NetInsight`
6. Use Mongoose to establish a connection when the backend starts.

Data Flow

- The user performs a network test in the mobile application.
- The application sends the results to the backend through the API.
- The backend validates and processes the request.
- The data is stored in MongoDB Atlas.
- The backend returns a response to the application.
- The application displays updated information to the user.



6.8 Testing and Verification

The implemented backend was tested by sending requests from the mobile application and verifying that records were successfully stored in MongoDB Atlas.

The following functionalities were verified:

- Successful insertion of network metrics.
- Retrieval of stored metrics.
- Computation of network statistics.

- Submission and retrieval of feedback.
- Update of feedback resolution status.
- Successful communication between frontend, backend, and database.

For networkmetrics:

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Data Explorer

My Queries | Data Modeling

CLUSTERS (1)

Search clusters

admin | local | sample_mflix | comments | embedded_movies | movies | sessions | theaters | users | test | feedbacks | **networkmetrics**

Cluster0 > test > networkmetrics

Documents 95 | Aggregations | Schema | Indexes 3 | Validation

Type a query: { field: 'value' } or [Generate query](#)

ADD DATA | BULK | EXPORT CODE

25 | 1 - 25 of 95

	LatencyStatus String	location Object	createdAt Date	updatedAt Date	_id Int32
1	"fair"	{ } 3 fields	2026-06-13T11:59:25.857+...	2026-06-13T11:59:25.857+...	0
2	"fair"	{ } 3 fields	2026-06-13T11:59:44.980+...	2026-06-13T11:59:44.980+...	0
3	"poor"	{ } 3 fields	2026-06-13T12:06:10.821+...	2026-06-13T12:06:10.821+...	0
4	"poor"	{ } 3 fields	2026-06-13T12:14:26.658+...	2026-06-13T12:14:26.658+...	0
5	"poor"	{ } 3 fields	2026-06-13T12:22:24.296+...	2026-06-13T12:22:24.296+...	0
6	"fair"	{ } 3 fields	2026-06-13T12:22:48.106+...	2026-06-13T12:22:48.106+...	0
7	"fair"	{ } 3 fields	2026-06-13T12:22:54.622+...	2026-06-13T12:22:54.622+...	0
8	"poor"	{ } 3 fields	2026-06-13T12:25:39.059+...	2026-06-13T12:25:39.059+...	0
9	"poor"	{ } 3 fields	2026-06-13T12:43:47.199+...	2026-06-13T12:43:47.199+...	0

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Data Explorer

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Cluster0 > test > networkmetrics

Documents 95 | Aggregations | Schema | Indexes 3 | Validation

Type a query: { field: 'value' } or [Generate query](#)

ADD DATA | BULK | EXPORT CODE

25 | 1 - 25 of 95

	_id ObjectId	deviceId String	timestamp Date	connectionType String	isConnected Boolean	ts
1	ObjectId('6a2d461d61d206...')	"device-ios"	2026-06-13T11:59:25.251+...	"cellular"	true	tr
2	ObjectId('6a2d463061d206...')	"device-ios"	2026-06-13T11:59:44.368+...	"cellular"	true	tr
3	ObjectId('6a2d47b261d206...')	"device-ios"	2026-06-13T12:06:10.896+...	"cellular"	true	fc
4	ObjectId('6a2d49a261d206...')	"device-ios"	2026-06-13T12:14:25.985+...	"cellular"	true	fc
5	ObjectId('6a2d4b8061d206...')	"device-ios"	2026-06-13T12:22:23.735+...	"cellular"	true	tr
6	ObjectId('6a2d4b9861d206...')	"device-ios"	2026-06-13T12:22:47.555+...	"cellular"	true	tr
7	ObjectId('6a2d4b9e61d206...')	"device-ios"	2026-06-13T12:22:54.851+...	"cellular"	true	tr
8	ObjectId('6a2d4c4361d206...')	"device-ios"	2026-06-13T12:25:38.501+...	"cellular"	true	fc
9	ObjectId('6a2d508361d206...')	"device-ios"	2026-06-13T12:43:46.591+...	"cellular"	true	fc

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My Queries

Data Modeling

CLUSTERS (1)

Search clusters

admin

local

sample_mfix

comments

embedded_movies

movies

sessions

theaters

users

test

feedbacks

networkmetrics

users

feedbacks

+

Cluster0 > test > feedbacks

View monitoring

Visualize Your Data

Documents 8

Aggregations

Schema

Indexes 2

Validation

Type a query: { field: 'value' } or [Generate query](#)

Explain

Reset

Find

Options

ADD DATA

BULK

EXPORT CODE

25

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	_id ObjectId	experience String	areaOfFeedback String	description String	rating Int32	loca
1	ObjectId('6a2d461861d206...	"fair"	"signal_strength"	"Not good at all."	3	{ } 2
2	ObjectId('6a2d464a61d206...	"fair"	"data_speed"	"It's all working well"	3	{ } 2
3	ObjectId('6a2d4ca161d206...	"fair"	"data_speed"	"Network is not that bad..."	3	{ } 2
4	ObjectId('6a2d556d61d206...	"fair"	"data_speed"	"Network is not good."	3	{ } 2
5	ObjectId('6a2e5c3061d206...	"excellent"	"customer_service"	"No"	5	{ } 2
6	ObjectId('6a2eaa9761d206...	"fair"	"data_speed"	"Not good at all."	3	{ } 2
7	ObjectId('6a2eb03561d206...	"poor"	"data_speed"	"Not good today"	2	{ } 2
8	ObjectId('6a2ec88461d206...	"excellent"	"data_speed"	"Good actually."	5	{ } 2

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My Queries

Data Modeling

CLUSTERS (1)

Search clusters

admin

local

sample_mfix

comments

embedded_movies

movies

sessions

theaters

users

test

feedbacks

networkmetrics

users

feedbacks

+

Cluster0 > test > feedbacks

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Explain

Reset

Find

Options

ADD DATA

BULK

EXPORT CODE

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	location Object	networkProvider String	resolved Boolean	createdAt Date	updatedAt Date	...
1	{ } 2 fields	"Anonymous"	false	2026-06-13T11:59:20.373+...	2026-06-13T11:59:20.373+...	0
2	{ } 2 fields	"Anonymous"	false	2026-06-13T12:00:10.085+...	2026-06-13T12:00:10.085+...	0
3	{ } 2 fields	"Anonymous"	false	2026-06-13T12:27:13.755+...	2026-06-13T12:27:13.755+...	0
4	{ } 2 fields	"Anonymous"	false	2026-06-13T13:04:45.956+...	2026-06-13T13:04:45.956+...	0
5	{ } 2 fields	"Anonymous"	false	2026-06-14T07:45:52.561+...	2026-06-14T07:45:52.561+...	0
6	{ } 2 fields	"Anonymous"	false	2026-06-14T13:20:23.985+...	2026-06-14T13:20:23.985+...	0
7	{ } 2 fields	"Anonymous"	false	2026-06-14T13:44:21.907+...	2026-06-14T13:44:21.907+...	0
8	{ } 2 fields	"Anonymous"	false	2026-06-14T15:28:04.077+...	2026-06-14T15:28:04.077+...	0

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6.9 Conclusion

Task 6 successfully implemented the database and backend components of the NetInsight system. The conceptual design and ER model provided a structured foundation for development, while MongoDB Atlas, Node.js, railway, Express.js, and Mongoose enabled efficient storage and management of network performance data and user feedback. The backend APIs were successfully connected to the database, allowing seamless communication between the mobile application and the cloud-hosted data store.